A SMART AI FRAMEWORK FOR AUTOMATED CIVIL COMPLAINT MANAGEMENT AND CITIZEN ENGAGEMENT IN SRI LANKA

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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ABSTRACT

E-governance systems play a vital role in improving communication between citizens and government institutions by enabling efficient handling of public complaints. However, in many developing countries, including Sri Lanka, complaint management processes are still largely manual, leading to delays, misclassification, and reduced service efficiency. Therefore, there is a need for an intelligent system that can automate complaint classification and routing while improving overall service delivery.

This research proposes GovConnect, a web-based government–citizen communication platform that integrates Artificial Intelligence techniques to enhance complaint management. The system utilizes Retrieval-Augmented Generation (RAG) combined with the LLaMA 3 Large Language Model (LLM) to classify citizen complaints into relevant government departments based on contextual understanding. A structured knowledge base was developed using data collected from government offices, including department descriptions, keywords, synonyms, and complaint examples.

The methodology includes data collection, preprocessing, embedding generation using nomic-embed-text, and classification through a hybrid approach combining semantic similarity, keyword matching, and a confidence threshold mechanism. The system also supports a complete workflow where complaints are routed to ministers and assigned to agents based on department and geographical location. The implementation was carried out using the MERN stack with JWT-based authentication and role-based access control.

Experimental testing was conducted using real-world complaint data, where the system achieved a classification accuracy of 87.18%, correctly classifying 34 out of 39 test cases. The results demonstrate that the proposed approach effectively handles unstructured complaint data and improves classification performance compared to traditional methods.

In conclusion, GovConnect provides a scalable and intelligent solution for automating complaint management, improving efficiency, accuracy, and transparency in government–citizen interactions.

Keywords: E-governance, Complaint Classification, RAG, LLaMA, Natural Language Processing

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TABLE OF ABBSERVATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
DB	Database
JWT	JSON Web Token
LLM	Large Language Model
MERN	MongoDB, Express.js, React.js, Node.js
ML	Machine Learning
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
UI	User Interface
UX	User Experience
HTTP	HyperText Transfer Protocol
JSON	JavaScript Object Notation
CRUD	Create, Read, Update, Delete
REST	Representational State Transfer

1. INTRODUCTION

E-governance has become an essential component in modern public administration as governments increasingly adopt digital technologies to improve communication with citizens and enhance service delivery. In many countries, citizen complaints serve as a primary medium through which the public reports issues, requests services, and expresses concerns related to various government departments. These complaints play a significant role in identifying real-world problems faced by society and help authorities take necessary actions to improve public services. However, despite the importance of complaint management, many existing systems are still dependent on manual or semi-automated processes, which limit their efficiency and scalability.

In traditional complaint handling systems, government officers are responsible for reading each complaint, identifying the relevant department, prioritizing the issue, and assigning it to responsible personnel. This process is time-consuming and highly dependent on human judgment, which may lead to inconsistencies, misclassification, and delays in response. As the number of complaints increases, managing them efficiently becomes a challenging task. In addition, complaints are usually submitted as unstructured text, making it difficult for rule-based systems to accurately interpret the content and determine the correct department. These limitations highlight the need for an intelligent and automated complaint management system that can improve accuracy, speed, and overall efficiency.

Recent advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), and Large Language Models (LLMs) provide effective solutions to address these challenges. NLP techniques enable systems to analyze and understand textual data by capturing contextual and semantic meaning rather than relying solely on keywords. Furthermore, Retrieval-Augmented Generation (RAG) has emerged as a powerful technique that

combines language models with external knowledge sources, improving the accuracy and reliability of predictions in domain-specific applications. By utilizing RAG, complaint classification systems can retrieve relevant contextual information such as department descriptions, keywords, and complaint examples, allowing more accurate mapping of complaints to appropriate government departments.

In addition to classification, effective complaint management requires prioritization and proper assignment of tasks. Machine learning algorithms can be used to estimate the urgency of complaints based on factors such as citizen-defined priority, public engagement, and contextual importance. This allows authorities to address critical issues more quickly and allocate resources efficiently. Moreover, integrating complaint classification and prioritization with administrative workflows such as minister-level review and agent assignment ensures that complaints are handled in a structured and transparent manner. Proper tracking of complaint status—from submission to resolution—further enhances accountability and trust between citizens and government authorities.

However, many existing research studies and systems focus only on complaint classification and do not provide a complete solution that integrates classification, prioritization, assignment, and monitoring into a single platform. In real-world government environments, complaints must be linked to departments, filtered by district, assigned to relevant agents, and continuously monitored until completion. The lack of such integrated systems creates inefficiencies and reduces the effectiveness of complaint handling processes.

Therefore, this research proposes **GovConnect**, an intelligent web-based government–citizen communication platform designed to enhance complaint management using advanced AI techniques. The system enables citizens to submit complaints through a user-friendly interface, automatically classifies them into relevant departments using a RAG-based approach with a fine-tuned LLaMA 3 model, and prioritizes them using a machine learning model. Subsequently, complaints are reviewed by the respective minister and

assigned to suitable agents based on department and district. Agents update the complaint status as pending, in progress, rejected, or completed, while both citizens and authorities can monitor the progress in real time.

The main contribution of GovConnect lies in integrating multiple intelligent components—complaint classification, urgency prediction, and agent assignment—into a unified e-governance platform. Unlike existing systems that address only isolated aspects, GovConnect provides an end-to-end solution for complaint handling, improving efficiency, accuracy, transparency, and accountability in public service delivery. As a result, this system has the potential to significantly enhance government–citizen communication and contribute to the development of smarter and more responsive governance systems

1.1. Background & Literature Review

Artificial Intelligence (AI) has become increasingly important in e-governance systems, especially in improving the way governments communicate with citizens and handle public complaints. In many countries, citizen complaint management is still carried out through manual or semi-digital processes, where officers read complaint descriptions, identify the relevant department, forward the complaint, and later monitor its progress. This process is often slow, inconsistent, and inefficient when the number of complaints becomes large. Delays in complaint classification and routing can reduce service quality and public trust. Therefore, intelligent complaint management systems are needed to automate complaint understanding, classification, prioritization, and assignment. In this research, a web-based government–citizen communication platform is proposed to address these issues. The system allows citizens to submit complaints online, automatically identifies the relevant government department using a Retrieval-Augmented Generation (RAG) system integrated with a fine-tuned Ollama LLaMA 3 model, prioritizes complaints using a machine learning model, and supports minister-level monitoring and agent assignment based on district and department relevance.

According to previous studies, many researchers have explored complaint classification using text mining, natural language processing, and machine learning techniques. Early studies proved that complaint reports can be automatically classified using text analysis methods. Sano, Yamaguchi, and Mine presented an automatic classification method for city park complaints and showed that text categorization can support municipal services by reducing manual work [1]. Fauzi proposed an automatic complaint classification system using classifier ensembles and demonstrated that combining multiple classifiers can improve performance and reliability [2]. Qurat-ul-ain, Shaukat, and Saif also introduced an NLP-based complaint classification model, confirming that textual complaint descriptions can be analyzed effectively for automated classification [3]. These works show that complaint text contains meaningful patterns that can be used to support decision-making in service systems.

Later, researchers extended these approaches using machine learning and deep learning models. Naik et al. studied consumer complaint classification using both machine learning and deep learning methods and found that automated systems can improve consistency and reduce human workload [4]. Saha et al. proposed an automated complaint classification and routing system using NLP and machine learning, highlighting that correct routing is as important as correct classification because complaints must reach the proper authority to be useful in practice [5]. However, many of these earlier methods depend on traditional feature engineering methods such as Bag-of-Words and TF-IDF, which are limited in understanding semantic meaning and contextual relationships in complaint text. Such approaches may perform poorly when complaints are short, ambiguous, or written in different styles.

With the advancement of deep learning, more powerful models have been introduced for text classification. Recurrent neural networks such as Long Short-Term Memory (LSTM) were developed to capture long-range dependencies in text and overcome limitations of standard recurrent models [11]. Later, the Transformer architecture introduced by Vaswani et al. significantly improved contextual text understanding through self-attention

mechanisms, enabling better modeling of semantic relationships across a sequence [12]. Fine-tuning large pretrained language models for text classification also became a major breakthrough. Howard and Ruder showed that language model fine-tuning can significantly improve classification performance even with limited labeled data [14]. In addition, broader survey studies confirm that deep learning methods outperform traditional machine learning techniques in many text classification tasks because they can automatically learn complex and meaningful representations from text [15].

Recent research has further advanced complaint classification by using embeddings and Large Language Models (LLMs). Rakhimzhanov, Belginova, and Yedilkhan proposed automated classification of public transport complaints using text mining with LLMs and embeddings, showing that modern semantic representations can improve complaint understanding in public service environments [6]. This is highly relevant to e-governance systems, where citizen complaints may contain multiple issues, indirect statements, and domain-specific wording. Compared to earlier methods, LLM-based approaches are more capable of understanding meaning beyond simple keyword matching.

Another important advancement in this area is Retrieval-Augmented Generation (RAG). Lewis et al. introduced RAG as a method that combines retrieval of relevant external knowledge with language model generation, improving performance on knowledge-intensive NLP tasks [7]. More recently, Adaptive-RAG demonstrated that retrieval strategies can be adjusted dynamically depending on the complexity of the input, leading to better efficiency and accuracy [8]. This is particularly useful for complaint classification because the same word may indicate different departments depending on the context. For example, terms like road, accident, hospital, garbage, or power cut may need contextual understanding before assigning the complaint to the correct department. By retrieving department descriptions, keywords, synonyms, and complaint examples from a structured knowledge base, a RAG-based model can produce more accurate and reliable department classification than a standalone text classifier.

Although previous studies have shown strong results for complaint classification, several limitations remain. Most existing research focuses mainly on classification accuracy and does not provide a complete workflow for government complaint handling. In real public service systems, complaint management involves not only identifying the correct department, but also prioritizing complaints based on urgency, forwarding them to relevant authorities, assigning them to field agents, and monitoring their progress until completion. Existing systems often do not fully integrate these operational requirements. In addition, many previous studies are not designed specifically for government complaint platforms where complaints must be linked to departments, ministers, districts, and agents in a structured administrative process.

Therefore, this research addresses an important gap by developing an integrated complaint management web application for government–citizen communication. In the proposed system, citizens submit complaints through the platform, the complaint is automatically classified into the relevant department using a RAG-based LLaMA 3 model, then prioritized using a RandomForestRegressor-based urgency model that considers complaint priority and public voting, and finally reviewed by the relevant minister who assigns the complaint to the most suitable agent in the corresponding district and department. Agents update complaint statuses such as pending, in progress, done, or rejected, while both citizens and ministers can monitor complaint progress through the system. Unlike many previous studies that focus only on classification, this research combines AI-based complaint classification, urgency prediction, and practical assignment workflows in one e-governance platform. Therefore, the proposed system contributes both academically and practically by moving from isolated text classification toward a complete intelligent complaint management solution for public administration.

1.2Research gap

In the field of e-governance and public service systems, despite numerous

research efforts in complaint classification, text mining, and Natural Language Processing (NLP) [1–6], the development of a fully automated and intelligent complaint management system still remains a challenging and partially unsolved problem. Citizen complaints are typically submitted as unstructured textual data with variations in writing style, length, grammar, and clarity. This diversity makes it difficult for existing systems to accurately interpret complaint content and map it to the correct government department. Although many studies have focused on improving classification accuracy, these solutions are often limited to controlled environments and fail to address real-world complexities in government applications.

In many existing government–citizen communication platforms, complaint classification and department assignment are still performed manually. Administrative officers are required to read each complaint and decide the relevant department based on their understanding. This process is time-consuming, inconsistent, and highly dependent on human judgment. As the number of complaints increases, manual classification becomes inefficient and leads to delays in service delivery, reduced responsiveness, and potential misclassification. Therefore, automating complaint classification is essential to improve efficiency and reliability in modern e-governance systems.

Earlier research has attempted to address this problem using traditional machine learning techniques such as Naïve Bayes, Support Vector Machines (SVM), Random Forest, and TF-IDF-based approaches [2]–[5]. While these methods improve processing speed compared to manual systems, they rely heavily on keyword matching and structured textual input. In real-world scenarios, citizens often submit complaints using short, incomplete, or informal sentences, which makes it difficult for these models to capture the actual meaning and intent of the complaint. As a result, traditional approaches fail to provide consistent and accurate classification, highlighting a significant limitation in handling real-world complaint data.

Recent advancements in deep learning and Large Language Models (LLMs) have significantly improved text understanding capabilities by capturing contextual and semantic relationships within text [6], [14], [15]. However,

when used independently, LLMs may lack domain-specific knowledge required for accurate classification in government environments. For instance, determining whether a complaint belongs to transport, health, or waste management requires an understanding of government responsibilities and contextual knowledge of real-world scenarios. Existing research shows limited application of Retrieval-Augmented Generation (RAG) techniques in complaint classification tasks, even though RAG enhances model performance by incorporating external knowledge sources [7], [8]. Therefore, there is a clear research gap in integrating LLMs with structured domain knowledge such as department descriptions, keywords, synonyms, and complaint examples to achieve more accurate and reliable classification.

Furthermore, most existing systems focus only on classification accuracy and do not provide intelligent department suggestion mechanisms during complaint submission. In practical government applications, it is important not only to classify complaints after submission but also to guide users by suggesting the correct department in real time. This reduces user confusion and ensures that complaints are routed correctly from the beginning. However, current systems lack such intelligent, context-aware suggestion mechanisms, especially for unstructured complaint text, which highlights another important research gap.

Another critical limitation in existing research is the lack of integration between complaint classification and government workflow processes. In real-world systems, once a complaint is classified into a department, it must be reviewed by the relevant minister and assigned to an appropriate agent based on department and district. However, most research focuses only on classification models without considering how these outputs are used in operational decision-making. There is limited research that integrates AI-based classification with agent assignment, complaint tracking, and status management systems. This creates a gap in developing a complete end-to-end complaint management solution that connects classification with real-world administrative processes.

In addition, government complaint systems require continuous monitoring

and efficient task distribution among agents. Existing systems provide limited support for filtering complaints based on department, location, and responsibility. Ministers often lack intelligent tools to assign complaints to the most suitable agents, resulting in inefficient workload distribution and delayed responses. Therefore, there is a need for systems that not only classify complaints but also support real-time monitoring, filtering, and intelligent assignment within the government structure.

Moreover, most existing studies are limited to experimental research conducted on benchmark datasets and do not provide fully functional implementations for real-world use. There is a lack of systems that integrate advanced AI techniques such as LLMs and RAG into a complete web-based platform for government–citizen communication. This creates a significant gap between theoretical research and practical application, particularly in developing countries where efficient complaint management systems are highly needed.

If the main research gap is summarized, it is the "Lack of an integrated, intelligent, and real-world e-governance platform that combines context-aware complaint classification using Large Language Models and Retrieval-Augmented Generation, automatic department suggestion, machine learning-based prioritization, and intelligent agent assignment into a single unified system."

This research gap is critically important because inefficient complaint handling directly affects public service quality, citizen satisfaction, and trust in government institutions. Misclassification, delays, and improper assignment can lead to unresolved issues and reduced system effectiveness. Therefore, addressing this gap through the development of an intelligent and fully integrated system such as GovConnect is essential to enhance the efficiency, accuracy, transparency, and accountability of government–citizen communication systems.

Reference	Functionalities of the Application	Technology and techniques Used for implementation	Platform/Usage	Accuracy
Traditional Government Complaint Systems	Manual complaint submission, manual classification, manual assignment	Human-based decision making	Web / Office-based	Low (time-consuming, error-prone)
Basic E-Government Portals	Online complaint submission, basic tracking	Rule-based systems, keyword matching	Web platforms	Low to Medium
Sano et al. [1]	Automatic classification of complaint reports	Text mining, basic NLP	Research prototype	Medium
Fauzi [2]	Complaint classification using ensemble classifiers	Machine Learning	Experimental system	Medium
Qurat-ul-ain et al. [3]	NLP-based complaint classification	NLP techniques	Research system	Medium
Naik et al. [4]	Consumer complaint classification	ML + Deep Learning models	Experimental datasets	Medium to High
Saha et al. [5]	Automated complaint classification and routing	NLP + Machine Learning	Prototype system	
Rakhimzhanov et al. [6]	Transport complaint classification using embeddings	LLM + Embeddings	Research system	High
Existing Smart Complaint Apps	Complaint submission and tracking only	Basic database + UI	Mobile / Web apps	Medium
LLM-based Systems (General)	Text understanding and classification	Transformers, LLMs	Research / APIs	High but lacks domain knowledge
RAG-based Systems [7][8]	Knowledge-enhanced text processing	LLM + Retrieval-Augmented Generation	Advanced research	High
GovConnect – Proposed System	Complaint submission, auto classification, department suggestion, minister review, agent assignment, real-time tracking	RAG + LLaMA 3 (LLM),	Web-based real-world system	Higher accuracy + full workflow integration

1.3. Research Problem

The problem regarding the efficient handling, classification, and management of citizen complaints in government systems should be addressed through an intelligent and automated approach. In many countries, including Sri Lanka, government–citizen communication mainly relies on complaint submission systems where citizens report issues related to public services such as transport, health, waste management, and electricity. However, these systems are still largely dependent on manual processes. When a citizen submits a complaint, administrative officers must read and analyze the complaint, identify the relevant department, prioritize the issue, and assign it to responsible personnel. This process is time-consuming, inefficient, and highly dependent on human judgment.

There are several challenges in the current complaint management process. Complaints are usually submitted as unstructured textual data, which varies in format, clarity, and detail. Citizens often provide incomplete, unclear, or informal descriptions of problems. As a result, it becomes difficult for human operators or rule-based systems to accurately understand the complaint and assign it to the correct department. Manual classification also leads to inconsistencies, misclassification, and delays in service delivery, especially when handling many complaints. This is one of the major problems in this domain.

Additionally, government officials such as administrators and ministers are required to monitor complaints and assign them to appropriate agents based on department and district. This process requires significant effort and coordination. In many cases, there is no intelligent support system to assist in selecting the most suitable agent for a given complaint. As a result, complaints may be assigned inefficiently, leading to delays in resolution and poor resource utilization. Furthermore, the lack of real-time monitoring and structured workflows makes it difficult to track the progress of complaints effectively.

Researchers in the fields of e-governance, artificial intelligence, and data

processing have attempted to address parts of this problem using machine learning and natural language processing techniques. However, most existing solutions focus only on complaint classification and do not consider the complete workflow, including prioritization, department suggestion, agent assignment, and status tracking. Moreover, traditional classification approaches struggle to understand the true meaning of complaints due to their reliance on keywords and structured inputs. This makes them less effective in real-world applications where complaint data is highly diverse and unstructured.

Another major issue is the lack of integration between intelligent classification systems and real-world government operations. In practical scenarios, complaints must not only be classified but also reviewed by ministers, assigned to agents in specific districts, and monitored until completion. The absence of such integrated systems results in fragmented processes and reduced efficiency in complaint handling. This highlights the need for a complete solution that connects automated classification with operational decision-making.

Furthermore, the increasing number of complaints submitted by citizens makes it difficult for existing systems to scale efficiently. Without automation, handling large volumes of complaints becomes a significant burden for government authorities. Delays in resolving complaints can reduce citizen satisfaction and weaken trust in government services. Therefore, there is an urgent need for a system that can process complaints quickly, accurately, and efficiently while supporting real-time monitoring and decision-making.

As a solution to these problems, this research proposes the development of GovConnect, an intelligent web-based government–citizen communication platform. The system utilizes advanced technologies such as Retrieval-Augmented Generation (RAG) and a fine-tuned Large Language Model (LLaMA 3) to automatically classify complaints into relevant departments. Additionally, a machine learning model is used to prioritize complaints based on factors such as user priority and public engagement. The system also supports minister-level monitoring and intelligent agent assignment based on

department and district, ensuring efficient complaint handling from submission to resolution.

Therefore, the main research problem can be summarized as the lack of an automated, intelligent, and integrated system for accurate complaint classification, prioritization, and agent assignment in government–citizen communication platforms, which results in inefficiency, delays, and reduced effectiveness in public service delivery.

1.4. Research Objectives

1.4.1. Main Objective

The primary objective of the proposed solution was to develop a web-based government–citizen communication platform, named GovConnect, that enables citizens to submit complaints and automatically classifies them into the relevant government departments using an intelligent and context-aware approach. The system was designed to utilize advanced Artificial Intelligence techniques, specifically Retrieval-Augmented Generation (RAG) integrated with a fine-tuned Large Language Model (LLaMA 3), to accurately understand and interpret unstructured complaint descriptions submitted by users. Through this platform, citizens can easily submit complaints without requiring prior knowledge of government structures or departments, while the system ensures that each complaint is correctly routed to the appropriate authority. Furthermore, the platform supports government workflow by allowing ministers to review categorized complaints and assign them to suitable agents based on department and district, enabling efficient task distribution and management. The system also provides complaint status tracking to enhance transparency and accountability. Therefore, the proposed solution aims to improve the efficiency, accuracy, scalability, and responsiveness of complaint handling processes, ultimately strengthening government–citizen interaction and contributing to more effective public service delivery.

1.4.2. Specific Objectives

1. To collect and prepare a structured knowledge base consisting of government departments, including descriptions, keywords, synonyms, and complaint examples obtained from relevant government sources.
2. To develop an intelligent complaint classification model using Retrieval-Augmented Generation (RAG) integrated with a fine-tuned LLaMA 3 model.

   • The RAG system retrieves relevant department-related information to improve the contextual understanding of complaints.
3. To analyze unstructured complaint text data submitted by citizens and accurately classify them into the appropriate government departments.

   • The model should be capable of understanding different writing styles and informal complaint descriptions.
4. To implement a web-based application using modern technologies (MERN stack with Vite and TailwindCSS) to provide a user-friendly interface for complaint submission and management.
5. To design and develop a system that allows ministers to view complaints filtered by their respective departments.
6. To develop an agent management and assignment mechanism that enables ministers to assign complaints to appropriate agents based on department and district.

   • The system should support filtering of agents according to complaint location and department relevance.
7. To implement a complaint status tracking mechanism where agents can update the progress of complaints (e.g., pending, in progress, completed, rejected).
8. To ensure real-time monitoring of complaint status by both citizens and government officials to improve transparency and accountability.
9. To evaluate the performance of the proposed system in terms of classification accuracy and practical usability in a real-world government environment.

2. METHODOLOGY

2.1. Understanding the Key Pillars of Research Domain

The proposed GovConnect system is fundamentally built upon several key technological and conceptual pillars, including Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), information retrieval techniques, web-based system engineering, role-based access control, and secure authentication mechanisms. These pillars collectively support the development of an intelligent complaint management system capable of automating the classification and routing of citizen complaints. The integration of these components ensures that the system is not only technically advanced but also capable of operating effectively within real-world government environments.

In traditional government complaint systems, classification and routing are performed manually, which leads to inefficiencies, inconsistencies, and delays. Therefore, the methodology of GovConnect is designed to address these challenges by combining text understanding, knowledge retrieval, and intelligent decision-making within a unified framework. This approach enables the system to interpret unstructured complaint text, identify the most relevant department, and integrate the result into a structured workflow involving ministers and agents. As a result, the system moves beyond simple classification and provides a complete end-to-end solution for complaint handling.

2.1.1. Natural Language processing

Natural Language Processing is a critical component of the GovConnect system because citizen complaints are typically submitted as free-text descriptions. In real-world scenarios, these complaints may be written in informal language, may contain spelling errors, and may lack clear structure. For example, a complaint such as “road broken vehicles cannot go” may not explicitly mention the department responsible for handling the issue, yet it clearly relates to transport or road maintenance.

To address this challenge, NLP techniques are used to extract meaningful information from complaint text. In the proposed system, complaint text is first preprocessed to remove unnecessary characters and normalize the content. It is then transformed into numerical representations using embedding techniques, allowing the system to analyze semantic meaning rather than relying solely on exact word matching. This enables the system to identify relationships between different complaint phrases and map them to appropriate government departments. The use of NLP significantly enhances the system's ability to handle real-world, unstructured input data and improves classification accuracy.

2.1.2. Retrieval-Augmented Generation

Retrieval-Augmented Generation represents the core innovation of this research and serves as the primary mechanism for improving classification performance. Traditional machine learning models rely solely on training data, while standalone LLMs may generate responses without sufficient domain grounding. In contrast, the RAG approach combines the strengths of both retrieval systems and generative models.

In the GovConnect system, a structured knowledge base is created using government department data, including descriptions, keywords, synonyms, and example complaints. This knowledge base is stored in a JSON format and converted into embeddings. When a complaint is submitted, the system retrieves the most relevant pieces of information from this knowledge base using similarity matching techniques. This retrieval step ensures that the classification process is grounded in domain-specific knowledge.

By providing the retrieved context to the language model, the system improves both accuracy and reliability. This approach reduces the risk of incorrect predictions and ensures that the classification aligns with real-world government responsibilities. Therefore, the use of RAG enhances both the interpretability and effectiveness of the system, making it more suitable for practical deployment.

2.1.3. Large Language Models

Large Language Models play a significant role in enabling advanced contextual understanding within the GovConnect system. In this research, the LLaMA 3 model is used through the Ollama framework to support intelligent decision-making. Unlike traditional models that rely on predefined rules or statistical patterns, LLMs can understand context, infer meaning, and make informed decisions based on input data.

However, instead of relying entirely on the LLM, the system adopts a hybrid approach. The LLM is used as a secondary decision layer that operates only when the retrieval confidence is insufficient. In such cases, the top candidate departments identified by the retrieval stage are provided to the LLM, and the model is prompted to select the most appropriate option. This controlled use of the LLM ensures that its output remains consistent and relevant to the domain.

This hybrid approach offers several advantages. It reduces unnecessary computational overhead, improves system reliability, and prevents incorrect or hallucinated responses. By combining the reasoning capabilities of LLMs with the precision of retrieval systems, the methodology achieves a balanced and efficient classification process.

2.1.4. Information Retrieval and Similarity Matching

Information retrieval techniques play a crucial role in the RAG-based classification process. The system uses embedding-based similarity matching to compare user complaints with stored department knowledge. Each complaint and knowledge entry is converted into a vector representation, allowing the system to measure similarity using mathematical distance metrics.

In addition to semantic similarity, the system also incorporates keyword-based scoring and synonym matching. This hybrid scoring mechanism ensures that both contextual meaning and explicit keywords are considered during classification. For example, a complaint mentioning “power cut” can be matched through both semantic similarity and keyword recognition, improving accuracy.

The use of a confidence threshold further enhances the reliability of the system. Only departments that meet the minimum threshold are considered valid suggestions, reducing the likelihood of incorrect classifications. This combination of retrieval techniques ensures that the system is both precise and robust.

2.1.5 Web-Based System Engineering

The GovConnect system is implemented as a web-based application to ensure accessibility, scalability, and ease of use. The system is developed using a modern technology stack, including React for the front end, Node.js and Express for the backend, and MongoDB for data storage. This architecture allows seamless interaction between users and the system while supporting real-time data processing.

The web-based design enables different user roles to interact with the system efficiently. Citizens can submit complaints through a user-friendly interface, ministers can manage and assign complaints, and agents can update complaint statuses. This integration ensures that the AI-based classification component is embedded within a functional and practical system rather than operating as a standalone model.

2.1.6. Role-Based Access Control and Security

Security and access control are essential components of the GovConnect system, especially because it handles sensitive complaint data and involves multiple user roles. The system implements role-based access control to ensure that users can only access functionalities relevant to their responsibilities.

Authentication is managed using JSON Web Tokens, which provide secure session management and protect API endpoints from unauthorized access. Each user role, including admin, citizen, minister, and agent, is assigned specific permissions within the system. This ensures data privacy, prevents misuse, and maintains system integrity.

By integrating security mechanisms into the system design, the methodology ensures that the platform is not only intelligent but also safe and reliable for

real-world deployment.

2.2. GovConnect Complaint Classification and Assignment Approach

The overall methodological approach of GovConnect was designed to develop a complete and practical complaint management workflow centered around intelligent department classification and operational assignment. The research began by identifying the main issue in current complaint systems: citizen complaints are handled manually, and this causes delays, inconsistency, and inefficient routing. To address this problem, the proposed methodology was organized into several major stages. First, complaint-related data were gathered from government offices. Second, these data were cleaned and transformed into a structured departmental knowledge base. Third, the knowledge base was embedded and indexed to support the RAG retrieval process. Fourth, an LLM-assisted classification mechanism was implemented to generate context-aware department suggestions. Fifth, the classified complaint was integrated into the application workflow so that ministers could review complaints and assign them to appropriate agents based on department and district. Finally, agents were enabled to update complaint statuses during the handling lifecycle. This phased approach ensures that the research solution is not only algorithmically sound but also operationally useful inside a web-based government platform.

2.2.1 Data Collection

In this research, the data used for the intelligent classification component were collected through visits to government offices and discussions with responsible officers. The data collection process included offices in Kaduwela, Athurugiriya, and Colombo municipal council environments, where complaint handling practices and departmental categories were discussed. Through these discussions, it was identified that the complaint management process is currently handled manually, where officers read each complaint and categorize it into relevant departments based on experience and human judgment.

According to the gathered information, the complaint data were mainly

categorized into departments such as Transport, Health, Environment & Public Nuisance, Waste Management, Education, Water Supply, and Electricity & Power. The officers also provided typical complaint examples and practical issue patterns related to each department. Based on these observations, a local complaint dataset was prepared using real-world complaint examples such as “Road is completely damaged and vehicles cannot pass,” “Hospital emergency ward overcrowded,” “Garbage not collected for two weeks,” “No water supply for three days,” and “Power cut every evening.” These complaint samples formed the foundation for the structured knowledge base used in the RAG system. Unlike generic public datasets, this research uses locally relevant government domain knowledge, which improves the practical validity of the proposed system.

2.2.2 Knowledge Base Preparation

After collecting the data, the next step was to prepare a structured knowledge base for the RAG system. The department collected data were organized into a file named knowledge.json. Each department entry contains a department name, a functional description, a list of keywords, a list of synonyms, and several complaint examples. For instance, the Electricity & Power department includes keywords such as *electricity, power, transformer, voltage, wire,* and *pole*, along with synonyms such as *power cut* and several practical complaint examples. This structure was selected because it allows the retrieval layer to access not only the department name but also multiple forms of supporting knowledge that help match a complaint more accurately.

The prepared structure improves complaint matching because a complaint may be related to a department through direct keywords, indirect synonyms, or similarity to past complaint examples. Therefore, the quality and organization of the knowledge base are critical to the accuracy of the RAG-based classification mechanism. In this research, the knowledge base acts as the domain memory of the system.

Example code structure used in knowledge.json

```
{
  "department": "Transport",
  "description": "Handles public transport services, road maintenance, traffic control, and vehicle licensing.",
  "keywords": ["road", "Bus","bus","traffic", "license", "highway", "bridge", "accident", "vehicle", "transport", "pavement", "pothole", "junction"],
  "synonyms": ["paare", "paara", "traffic jam", "රථවාහන", "බස්", "මාර්ග"],
  "complaintExamples": [
    "Road is completely damaged and vehicles cannot pass",
    "Bus service is not operating on time",
    "Heavy traffic jam every morning",
    "Illegal parking blocking the main road",
    "Broken bridge causing accidents",
    "Pothole on main road causing accidents",
    "Bus route cancelled without notice",
    "Traffic lights not working at junction",
    "Road is flooded and blocked"
  ]
},
```

Figure 1:Example code structure used in knowledge.json

2.2.3 Embedding Generation and Index Building

Once the knowledge base was prepared, it was transformed into an indexed retrieval resource using text embeddings. In the proposed implementation, each department entry was expanded into several textual chunks, including the department description, complaint examples, keyword summaries, and synonym summaries. These text chunks were then passed to the nomic-embed-text embedding model through Ollama's embedding API, and the resulting vector embeddings were stored in index.json. This process was implemented in the backend RAG module.

The purpose of this step is to convert symbolic department knowledge into vector representations so that complaint queries can be compared semantically rather than only through direct string matching. Through this process, the system becomes capable of matching a complaint to conceptually similar department content even when the wording is different from the stored examples.

```
import axios from "axios";

const OLLAMA_URL = "http://localhost:11434/api/embeddings";
const EMBED_MODEL = "nomic-embed-text";

export const getEmbedding = async (text) => {
  const response = await axios.post(OLLAMA_URL, {
    model: EMBED_MODEL,
    prompt: text,
  });
  return response.data.embedding;
};

```

Figure2.2.2: Example code for embedding

```
29
30      const documents = [
31        `Department: ${entry.department}. ${entry.description}`,
32        ...entry.complaintExamples.map((ex) => `${entry.department} complaint: ${ex}`),
33        `${entry.department} handles issues related to: ${entry.keywords.join(", ")}`,
34        ...(entry.synonyms ? [`${entry.department} also known as: ${entry.synonyms.join(", ")}`] : []),
35      ];
36
```

Figure 2.2.3: Example code for index building

2.2.4. Retrieval Process

After building the vector index, the retrieval stage was implemented. When a citizen enters a complaint title, the system generates an embedding for that complaint and compares it against the stored department chunk embeddings using cosine similarity. In parallel, the complaint text is also checked against department keywords, synonyms, and complaint examples to compute an additional keyword-based relevance boost. A final hybrid score is then calculated using both vector similarity and keyword relevance.

Departments are ranked by this hybrid score, and the top departments above a minimum confidence threshold are selected as retrieval results. In the proposed implementation, the minimum confidence threshold is set as **0.45**, which means that only departments with a hybrid score of at least 0.45 are accepted as valid suggestions. This helps improve reliability and avoids low-confidence classifications.

```
9
10    let _index = null;
11    const MIN_CONFIDENCE = 0.45; // min hybrid score to accept suggestion
12
```

```
24
25  // Keyword / synonym / example scoring
26  const keywordScore = (entry, query) => {
27    const q = query.toLowerCase();
28    let score = 0;
29
30    for (const kw of entry.keywords || []) if (q.includes(kw.toLowerCase())) score += 2;
31    for (const syn of entry.synonyms || []) if (q.includes(syn.toLowerCase())) score += 3;
32    for (const ex of entry.complaintExamples || []) {
33      const exWords = ex.toLowerCase().split(" ");
34      const qWords = q.split(" ");
35      const shared = qWords.filter((w) => w.length > 3 && exWords.includes(w)).length;
36      score += shared * 0.5;
37    }
38    return score;
39  };
40
```

```
const hybridScore = maxVectorSim * 0.6 + kwBoost * 0.4;
```

Figure 2.2.4: Example code for retriever

This retrieval process is a key methodological contribution because it improves department selection by combining semantic and lexical signals rather than depending only on exact keyword matches.

2.2.5 LLM-Based Department Suggestion

The retrieved department information is then used inside the LLM-based classification stage. If the retrieval system produces a highly confident top result, for example a hybrid scores greater than 0.85, the system directly uses that department. However, when the top result is not sufficiently confident, the complaint title and the top retrieved department options are passed to the LLaMA3 model through the Ollama generate API. The model is instructed to choose the best department from the retrieved options and return only the department name.

The system also validates the LLM's answer against the retrieved department list before using it. If the LLM response is empty or invalid, the system falls back to the highest-ranked department retrieved. This methodology ensures both robustness and efficiency because the LLM is used only when necessary and always inside a constrained and domain-aware decision space.

```
    // 🔥 Step 3: High confidence → skip LLM
    if (topDepts[0].hybridScore > 0.85) {
      return { department: topDepts[0].department };
    }

    // 🔥 Step 4: LLM fallback
    const prompt = `You are a Sri Lankan complaint routing assistant.

Choose the best department.

Options:
${topDepts.map((d) => d.department).join(", ")}

Complaint: "${complaintTitle}"

Answer ONLY the department name.`;

    const response = await axios.post(OLLAMA_URL, {
      model: OLLAMA_MODEL,
      prompt,
      stream: false,
    });
```

2.2.6 Complaint Submission and Department Routing

The complaint submission process was integrated into the citizen workflow of GovConnect. When the citizen submits a complaint, the complaint can either be routed using a manually selected department or can be automatically suggested by the AI component. In the implementation, if the department is not manually selected, the system invokes the RAG-based suggestion workflow, retrieves the most relevant department, matches it with the department records stored in the database, and stores the final department identifier with the complaint.

This stage connects the AI classification component to the actual complaint management logic of the application. Therefore, the methodology does not stop at model prediction; it continues into complaint creation and routing inside the database.

```
70
71    // Auto-suggest department if not selected manually
72    if (!departmentId) {
73      try {
74        const allDepartments = await Department.find({});
75        const suggestion = await buildRagContext(title); // 🔥 use RAG
76        const matched = matchDepartment(allDepartments, suggestion?.topDepts?.[0]?.department);
77
78        if (matched) departmentId = matched._id;
79        // If no match, leave null → frontend prompts manual selection
80      } catch (err) {
81        console.warn("RAG fallback failed during addComplaint:", err.message);
82      }
83    }
84
85    const newComplaint = new Complaint({
86      citizen: citizen._id,
87      name: citizen.name,
88      email: citizen.email,
89      title,
90      description,
91      phoneNumber,
92      address,
93      district,
94      department: departmentId,
95      priority,
96    });
97
```

Example code from complaint creation

2.2.7 Minister Review and Agent Assignment

After classification, complaints become visible to the minister responsible for the assigned department. The minister's dashboard provides access to complaints belonging to that department, and the minister can review them and assign them to appropriate field agents. For practical complaint handling, the minister must be able to assign complaints based on both department and

district.

For example, if a citizen from Matara district submits a complaint that is classified under the Transport department, the minister can retrieve agents relevant to that department and district, such as Road Service or Public Transport agents, and assign the complaint accordingly. This ensures that the AI output is embedded into a real operational chain rather than remaining as an isolated classification result.

```
// Get agents filtered by department and district
export const getAvailableAgents = async (req, res) => {
  const { department, district } = req.query;
  try {
    const filter = {};
    if (department) filter.department = department;
    if (district)   filter.district   = district;

    const agents = await Agent.find(filter).populate("department", "name");
    res.json(agents);
  } catch (error) {
    res.status(500).json({ error: "Server Error" });
  }
};
```

Example code for minister-side assignment

```

// Assign agent to complaint
export const assignAgentToComplaint = async (req, res) => {
  const { complaintId, agentId } = req.body;
  try {
    const minister = await Minister.findById(req.user._id);

    const complaint = await Complaint.findOne({
      _id: complaintId,
      department: minister.department, // minister can only assign within their dept
    });

    if (!complaint) {
      return res.status(404).json({ error: "Complaint not found in your department" });
    }

    complaint.assignedAgent = agentId;
    complaint.status = "IN_PROGRESS";
    await complaint.save();

    const updated = await Complaint.findById(complaintId)
      .populate("assignedAgent", "name email district agentType")
      .populate("department", "name")
      .populate("citizen", "name email");

    res.json({ message: "Agent assigned successfully", complaint: updated });
  } catch (error) {
    res.status(500).json({ error: "Server Error" });
  }
};
```

Example code for minister-side assignment

2.2.8 Agent Handling and Status Tracking

Once the minister assigns a complaint, the assigned agent becomes responsible for handling it. Agents can view complaints assigned to them and update the complaint status according to progress, such as IN_PROGRESS, RESOLVED, or REJECTED. This status-tracking mechanism is part of the methodology because the research is implemented as a complete end-to-end complaint handling platform. The classification result therefore becomes actionable, traceable, and monitorable.

Citizens and ministers can later view these status changes through their respective interfaces, which improves transparency and accountability in the government–citizen communication process.

```
// 👇 Agent updates complaint status
export const updateAgentComplaintStatus = async (req, res) => {
  const { status } = req.body;
  const validStatuses = ["IN_PROGRESS", "RESOLVED", "REJECTED"];

  if (!validStatuses.includes(status)) {
    return res.status(400).json({ error: "Invalid status" });
  }

  try {
    const complaint = await Complaint.findOne({
      _id: req.params.id,
      assignedAgent: req.user._id, // agent can only update their own assigned complaints
    });

    if (!complaint) return res.status(404).json({ error: "Complaint not found or not assigned to you" });

    complaint.status = status;
    await complaint.save();
    res.json({ message: "Status updated", complaint });
  } catch (error) {
    res.status(500).json({ error: "Server Error" });
  }
};
```

Example code for agent status update

2.2.9 Code Implementation of the Research Component

The implementation of the GovConnect research component is developed using modular backend architecture that integrates Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs) to perform intelligent complaint classification and routing. The implementation is structured into multiple components, including knowledge base preparation, embedding generation, retrieval-based similarity matching, LLM-assisted

decision making, and workflow integration. Each component is designed as an independent module to ensure maintainability, scalability, and clarity in system design.

The implementation begins with the preparation of the knowledge base, which is stored in a file named *knowledge.json*. This file contains structured information about government departments, including their descriptions, keywords, synonyms, and example complaints. This structured dataset acts as the domain knowledge source for the RAG system and enables the system to match user complaints using multiple contextual dimensions rather than relying solely on simple keyword matching.

Following this, the index-building process is implemented using the *buildIndex.js* module. In this stage, the knowledge base is transformed into multiple natural language text documents and converted into vector embeddings using the embedding model. These embeddings are stored in an index file (*index.json*), allowing efficient similarity-based retrieval during runtime. This transformation is essential because embedding models operate on textual input, and structured JSON data must be converted into descriptive sentences to capture semantic meaning effectively.

The embedding generation and similarity computation are handled through a dedicated embedding module, which communicates with the Ollama embedding API to generate vector representations for both stored knowledge and incoming complaint text. This module also performs cosine similarity calculations to determine the relevance between complaints and department data, enabling semantic matching instead of exact word comparison.

The retrieval logic is implemented in the *retriever.js* module, which identifies the most relevant department for a given complaint. When a complaint is submitted, the system generates an embedding for the complaint text and compares it with the stored embedding in the index. In addition to semantic similarity, keyword and synonym matching are applied to improve classification accuracy. A hybrid scoring mechanism is used to combine these factors, and departments are ranked based on their relevance scores. A predefined confidence threshold is applied to filter out low-confidence results,

ensuring reliable classification.

When the retrieval stage does not produce a highly confident result, the system invokes the LLM-based decision module implemented in *ollama.*service.js. In this stage, the top candidate departments identified during retrieval are provided to the LLaMA 3 model along with the complaint text. The model selects the most appropriate department from the given options, ensuring that the decision is context-aware and aligned with domain knowledge. The output is validated to ensure correctness and reliability.

The classification system is integrated into the application workflow through the complaint controller. When a citizen submits a complaint without selecting a department, the system automatically invokes the RAG-based classification process to determine the appropriate department. The selected department is then matched with database records and assigned to the complaint before storing it. This ensures that AI-based classification is directly integrated into the complaint creation process.

Once the complaint is created, the system proceeds to the operational workflow, where ministers and agents handle the complaint. The minister controller is responsible for reviewing department-specific complaints and assigning them to appropriate agents based on department and district. The agent controller enables agents to view assigned complaints and update their status throughout the process. This ensures that classified complaints are effectively managed within the system.

Overall, the modular implementation ensures a clear separation of concerns between knowledge management, embedding generation, retrieval logic, LLM interaction, and workflow integration. This design improves system maintainability, scalability, and performance, making the GovConnect system suitable for real-world government complaint management.

2.3. System Architecture Diagram

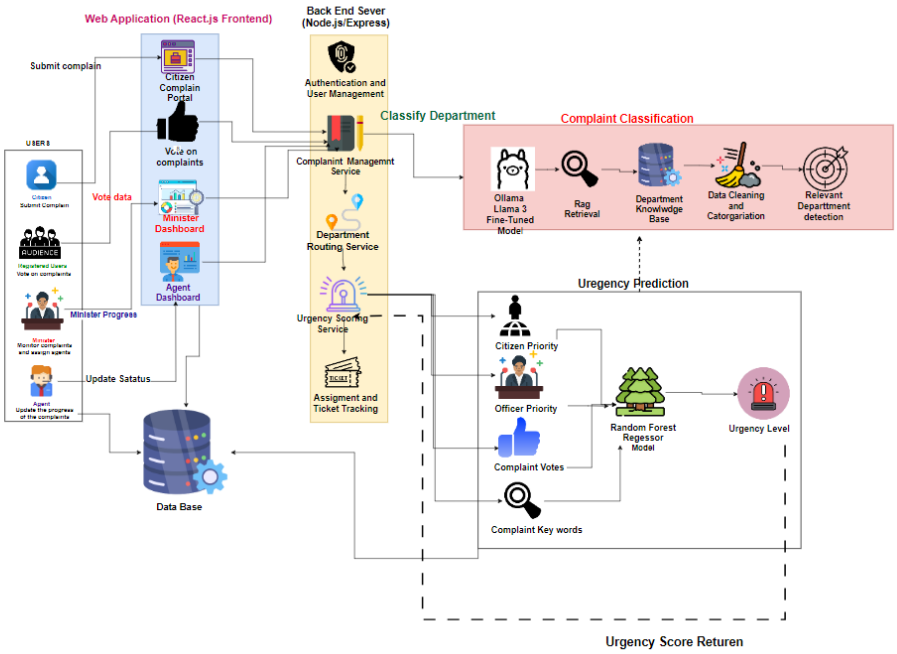


Figure 2.3.1: System Architecture Diagram

The system architecture of the GovConnect platform successfully integrates several elements that cooperate to automate the government-citizen complaint handling procedure. The architecture is divided into discrete levels, each of which is in charge of managing particular functions, guaranteeing an effective information flow, and facilitating smooth communication between users and governmentagencies.

React.js is used in the platform's frontend development, offering a dynamic and interactive interface for users to file complaints. While ministries and agents have access to dashboards to monitor and address problems, citizens can readily file complaints. Details from complaints submitted by citizens are processed in the backend for routing and classification.Assigning complaints to the proper agents and monitoring their progress are made possible by the Minister function. Agents are in charge of managing the complaints that are allocated to them, keeping their statuses up to date, and resolving problems.All essential functions are managed by the backend server, which is powered by Node.js and Express.js. This involves handling complaints and

utilizing JWT-based user management for authentication. Additionally, the backend incorporates RAG and LLaMA 3 models for complaint classification, automatically allocating complaints to the relevant department according to the substance of the complaint. In order to handle complaints according to urgency, the Urgency Scoring Service also use Random Forest Regressor to determine each complaint's priority.

Classification and urgency prediction heavily rely on AI models. By examining the complaint content and comparing it with department records kept in the knowledge base, RAG and LLaMA 3 collaborate to guarantee appropriate department classification. Additionally, the system uses Random Forest Regressor to forecast urgency while accounting for user votes, officer priority, and citizen priority.Lastly, complaint records, user profiles, departmental data, and agent assignments are stored in the database, which is run by MongoDB. This ensures that the system runs smoothly even as the number of complaints increases by enabling effective data retrieval and handling.

All things considered, the architecture of the GovConnect platform guarantees that complaints are resolved quickly, effectively, and precisely, minimizing manual labor and improving the citizen-government engagement process.

2.4. Functional and Non-Functional Requirements

2.4.1. Functional Requirements

Functional Name	Input	Processing	Output
Citizen Registration and Login	User details such as name, email, password,	System validates input, encrypts password, stores user details, and authenticates using JWT	Secure user account creation and login access
Complaint Submission	Complaint title, description, phone number,	System validates complaint data and records it in	Complaint stored successfully in the system

	address, district, priority, and optional department	the database	
AI Department Suggestion	Complaint title or complaint text entered by citizen	RAG system retrieves relevant knowledge from knowledge.json, compares embeddings, applies hybrid score, and selects the most suitable department	Suggested department returned to the citizen
Manual Department Selection (Optional)	Department selected manually by citizen	System bypasses AI suggestion and directly uses selected department	Complaint routed to selected department
Knowledge Base Matching	Complaint text and department knowledge data (description, keywords, synonyms, examples)	System checks semantic similarity and keyword relevance against indexed knowledge	Top matching departments identified
Complaint Storage and Routing	Complaint data with final department	System stores complaint in MongoDB under the matched department	Complaint available in relevant department workflow
Minister's Complaint View	Department-based complaint request from minister	System filters complaints according to the	Department complaint list displayed in

		minister's assigned department	minister dashboard
Agent Filtering and Assignment	Complaint department and district, available agent data	System filters agents by department and district; minister selects one for assignment	Complaint assigned to the most suitable agent
Agent Complaint Handling	Assigned complaint and updated status	Agent reviews complaint and updates status such as Pending, In Progress, Resolved, or Rejected	Updated complaint status stored in database
Complaint Tracking	Complaint ID and user role	System retrieves complaints, progress and status	Citizens, ministers, and agents can monitor complaint progress
Admin User Management	Minister, agent, and department details	Admin creates departments, registers ministers and agents, and assigns ministers to departments	Updated user and department records
Role-Based Access Control	Authenticated user token and role	System checks user role permissions before granting access to APIs and dashboards	Controlled access for admin, citizen, minister, and agent

Table 2.4.1: Functional Requirements

2.4.2. Non-Functional Requirements

NFR	Testable NFR
Performance	The system should return department suggestions within a few seconds under normal load.
Scalability	The system should support a growing number of complaints, users, departments, ministers, and agents without major performance degradation.
Availability	The system should remain accessible during normal working hours with minimum service interruption.
Reliability	The complaint classification and routing process should produce consistent results with minimal system failure.
Usability	A citizen should be able to submit a complaint with minimal effort using a simple and understandable interface.
Maintainability	The modular architecture should allow new departments, keywords, synonyms, and complaint examples to be added without major code restructuring.
Compatibility	The web application should work correctly on common modern browsers such as Chrome, Firefox, Edge, and Safari.
Security	The system should enforce JWT-based authentication and role-based access control to prevent unauthorized access.
Accuracy	The department classification component should maintain high accuracy using the RAG + LLM hybrid method and should correctly classify most real complaint cases.

Auditability	The system should preserve complaint records, assignments, and status updates so they can be reviewed when necessary.
Data Integrity	Complaint, user, and department data should be stored consistently without duplication or corruption.
Flexibility	The system should support future extensions such as adding new complaint categories, departments, and workflow features.

Table 2.4.2: Non-Functional Requirements

2.5. Social, Security, and Ethical Considerations

2.5.1. Social Impact

The GovConnect system plays a significant role in improving communication between citizens and government authorities by providing a centralized digital platform for complaint management. In many traditional systems, citizens face delays and uncertainty when reporting issues, as complaints are processed manually and often misrouted. By introducing an intelligent classification mechanism, the proposed system ensures that complaints are directed to the correct department at the initial stage, thereby reducing delays and improving response time.

Furthermore, the system enhances transparency in government operations by allowing citizens to track the progress of their complaints in real time. This visibility increases public trust and confidence in government services, as users are no longer required to rely on manual follow-ups or personal connections to receive updates. The inclusion of a voting mechanism also empowers citizens by allowing them to express the urgency of issues, ensuring that high-impact problems receive greater attention.

In addition, the system promotes inclusivity by providing a simple and user-friendly interface that can be used by individuals with varying levels of technical knowledge. This ensures that a wider population can access government services, contributing to the development of a more responsive

and citizen-centric governance model.

2.5.2. Security Considerations

Security is a critical component of the GovConnect system due to the sensitive nature of user data and complaint information. The system implements robust authentication and authorization mechanisms to ensure that only authorized users can access system functionalities. JSON Web Token (JWT) based authentication is used to manage user sessions securely, preventing unauthorized access and ensuring data confidentiality.

Role-Based Access Control (RBAC) is applied to restrict system functionalities based on user roles such as citizen, minister, agent, and administrator. This ensures that each user can only perform actions relevant to their responsibilities, reducing the risk of data misuse or unauthorized modifications. For example, only ministers can assign agents, and only agents can update complaint statuses.

To further enhance security, sensitive data such as user passwords are stored using hashing techniques, ensuring that they cannot be easily accessed or misused even in the event of a data breach. API endpoints are protected using authentication middleware, which verifies user identity before allowing access to backend services.

Additionally, input validation and sanitization techniques are applied to prevent common security threats such as injection attacks and malformed data submissions. These measures ensure that the system remains secure, reliable, and resistant to malicious activities.

2.5.3. Ethical Considerations

The use of Artificial Intelligence in complaint classification introduces several ethical considerations that must be carefully addressed. One of the primary concerns is the fairness and accuracy of the classification process. Since incorrect classification may lead to delays in resolving critical issues, the system is designed to include a fallback mechanism that allows users to manually select a department if the automated suggestion is not appropriate.

This ensures that users retain control over the final decision.

Another important ethical aspect is the prevention of bias in the classification model. The system relies on a structured knowledge base that represents multiple departments equally, ensuring that no department is unfairly prioritized over others. By using a RAG-based approach, the system grounds its decisions in predefined domain knowledge rather than relying solely on generative outputs, which helps reduce bias and improve reliability.

User privacy is also a major ethical concern. The system ensures that personal information such as names, contact details, and complaint descriptions are only accessible to authorized users. Data access is restricted based on user roles, and sensitive information is handled securely to prevent unauthorized exposure.

Furthermore, the system includes mechanisms to prevent misuse, such as restricting duplicate voting and ensuring accountability in complaint handling. Each action within the system, including complaint submission, assignment, and status updates, is recorded, enabling traceability and auditability. This promotes responsible use of the system and ensures that all users act within ethical boundaries.

Overall, the GovConnect system is designed with a strong emphasis on ethical responsibility, ensuring fairness, transparency, privacy, and accountability in all aspects of its operation.

2.6. Testing Strategy

From all the data collected through government office visits and real complaint scenarios, after the processes of data preparation, structuring into departments, and constructing the knowledge base (*knowledge.json*), the dataset was used to evaluate the performance of the GovConnect complaint classification system. Unlike traditional image-based classification models, the proposed system operates on textual complaint data, where each complaint is mapped to one of the predefined government departments using a Retrieval-Augmented Generation (RAG) approach combined with a Large Language Model (LLM).

For the purpose of evaluation, the prepared dataset was conceptually divided into two main components: the knowledge base dataset and the test dataset. The knowledge base dataset contains structured departmental information, including descriptions, keywords, synonyms, and complaint examples. This dataset acts as the reference knowledge source for the RAG system and is used to retrieve relevant contextual information during classification. It plays a role similar to a training dataset in traditional machine learning systems, although no explicit model training is performed.

The test dataset consists of unseen complaint examples that are not included in the knowledge base. These complaints represent real-world government issues and are used to evaluate the classification capability of the system. By ensuring that the test cases are independent from the knowledge base, the evaluation remains unbiased and reflects the actual performance of the system in practical scenarios.

In total, a set of 39 complaint test cases was used for evaluation. These test cases include complaints from seven major government departments, namely Transport, Health, Environment & Public Nuisance, Waste Management, Education, Water Supply, and Electricity & Power. In addition, a small number of unrelated inputs were included to test the robustness of the system in handling irrelevant or non-government complaints.

Complaint Class	**Test**
Transport	3
Health	5
Environment & Public Nuisance	5
Waste Management	5
Education	5
Water Supply	5
Electricity & Power	5
Non-relevant Inputs	6
Total	39

2.6.1: Class distribution of complaint test cases

The testing process evaluates the ability of the system to correctly classify complaints into the appropriate department using the RAG-based classification pipeline. When a citizen submits a complaint and selects the AI suggestion option, the complaint text is first processed and converted into an embedding vector using the embedding model. This embedding is then compared with the embeddings generated from the knowledge base using semantic similarity measures.

In addition to semantic similarity, the system applies keyword matching and synonym matching techniques to improve classification performance. A hybrid scoring mechanism is used to combine semantic similarity and keyword-based relevance. Based on this hybrid score, departments are ranked according to their relevance to the complaint.

To ensure reliable classification, a minimum confidence threshold of **0.45** is applied. Only departments with a hybrid score greater than or equal to this threshold are considered valid suggestions. If the top-ranked department satisfies this condition and is available in the system database, it is selected as the final classification result.

In cases where the hybrid score does not meet the required confidence level, the system invokes the LLaMA 3 model through the Ollama framework to assist in selecting the most appropriate department. The model is provided with the complaint text along with the top candidate departments retrieved during the previous stage. Based on contextual understanding, LLM selects the most suitable department. This hybrid RAG + LLM approach ensures both accuracy and robustness in handling ambiguous complaint inputs.

After the department is identified, the complaint is submitted into the system and routed to the corresponding department. The complaint then becomes visible in the relevant minister's dashboard, where the minister reviews the complaint and assigns it to an appropriate agent. The assignment process is based on both the department and the district of the complaint. For example, if a complaint is submitted from the Matara district under the Transport

department, the system allows the minister to assign the complaint to a relevant Road Service or Public Transport agent within that district. This ensures efficient handling and proper resource allocation.

In addition to classification and routing, the testing process also verifies the system's role-based access control and authentication mechanisms. The system ensures that only administrators can register ministers and agents, while citizens can submit complaints and access system functionalities based on their permissions. Authentication is implemented using JWT token-based login, which ensures secure access to system resources.

Overall, the testing methodology confirms that the GovConnect system effectively integrates AI-based complaint classification with real-world government workflows. The use of a structured knowledge base, embedding-based similarity matching, hybrid scoring mechanism, confidence thresholding, and LLM-assisted decision-making enables the system to handle a wide range of complaint inputs in a reliable and efficient manner.

2.7. Tools and Technologies Used

The GovConnect system was developed using a combination of modern web technologies, database tools, and Artificial Intelligence frameworks. These technologies were selected to support efficient system development, secure user management, intelligent complaint classification, and a responsive user experience. The selection of tools also ensured compatibility between different system modules and provided a scalable foundation for future improvements.

Frontend Technologies

The frontend of the GovConnect system was developed using React.js together with Vite and Tailwind CSS. React.js was selected because it supports component-based user interface development and allows dynamic rendering of content based on user roles and system data. Vite was used as the frontend build tool because it provides fast development performance and an efficient project structure. Tailwind CSS was used to design responsive and user-friendly interfaces, enabling clean styling and consistent visual

layouts across the citizen, minister, agent, and administrator pages. Axios was used in the frontend to communicate with backend APIs and retrieve or send data between the client and server.

Backend Technologies

The backend of the system was implemented using Node.js and Express.js. Node.js was selected because it provides an efficient runtime environment for server-side JavaScript and supports asynchronous operations, which are useful for handling multiple requests efficiently. Express.js was used to build the REST API structure of the system, including routes for complaint submission, authentication, assignment, and status updates. MongoDB was used as the database management system, and Mongoose was used as the object data modeling library. This combination enabled flexible storage of complaint records, user details, department information, and assignment data.

AI and Machine Learning Technologies

The intelligent complaint classification component of GovConnect was implemented using Ollama with the LLaMA 3 model and the nomic-embed-text embedding model. LLaMA 3 was used to provide context-aware reasoning when the retrieval layer did not produce a highly confident result. The nomic-embed-text model was used to generate embeddings for the structured departmental knowledge and incoming complaint text. The overall AI methodology was based on Retrieval-Augmented Generation, where relevant department information is first retrieved from the knowledge base and then used to support the final classification decision. This combination of retrieval and language model reasoning forms the main novelty of the proposed system.

Development Tools

Several development tools were used during implementation. Visual Studio Code was used as the primary source code editor because it provides a flexible environment for both frontend and backend development. Git and GitHub were used for version control and source code management, enabling organized development and easier tracking of system changes. Postman was used for API testing to verify backend endpoints and ensure correct request-

response behavior during development. These tools contributed to more efficient implementation, testing, and debugging of the system.

Authentication and Security Technologies

To ensure secure system access, JSON Web Tokens were used for authentication. JWT enabled secure login sessions and ensured that only authenticated users could access protected features. In addition, Role-Based Access Control was implemented to restrict access according to user roles such as citizen, minister, agent, and administrator. This ensured that each user could only access the functionalities relevant to their responsibilities, thereby improving system security and maintaining proper workflow control.

Deployment Environment

At the current stage, the system is mainly designed and tested in a local development environment. The frontend and backend are executed locally while the MongoDB database supports complaint and user data storage. Although large-scale cloud deployment is considered future work, the current implementation provides a strong foundation for deployment in a production environment. The modular structure of the system allows it to be extended and hosted on cloud platforms in future enhancements

4.RESULT AND DISCUSSION

4.1. Result

The GovConnect system was extensively tested using a battery of functional, real-time and system-level tests for their validity in automating government-citizen complaint processing. Built on the MERN (MongoDB, Express.js, React.js, Node.js) stack and augmented with Artificial Intelligence (RAG & LLaMA 3), the system is a comprehensive and robust system. The evaluation was performed on all important components like administration modules, user authentication, complaint classification via AI and all workflows including ministers and agents, and is successful. All the main components of the system are working correctly and end to end complaint management lifecycle

is supported.

4.1.1. AI-Based Complaint Classification

GovConnect contains its primary intelligent component, the AI-based classification module connected through the LLaMA 3 large language model to provide the context-aware classification of citizen complaints by accessing the Retrieval-Augmented Generation (RAG). A real-time assessment of this component was performed through execution on the backend system (see an example of a successful execution in the accompanying screenshot). The classification pipeline works appropriately and produces results by integrating semantic retrieval strategies with contextual reasoning.

RAG System Execution Results

As shown in the system execution output, when the complaint text:

"No medicine available"

was submitted, the system processed the input through the RAG pipeline and generated the following output:

The semantic similarity based embeddings produced have been shown to successfully produce comparable representations of the Complaint compared to previous calculated vector representations in our Knowledge Base (KB). The semantic match was found to be 0.578 (maximum value) between the Complaint and the KB indicating there is a significant semantic similarity between the complainant's health-related nature and the data contained in KB regarding all aspects of health. Therefore, due to the strong similarity between the two, the system has been able to very accurately identify the most appropriate department as having a corresponding semantic similarity and has demonstrated effectiveness of using an embedding approach for helping to

```
[RAG] Index loaded: 8 departments
[RAG] Top matches for "No medicine available": [ 'Health (0.578)' ]
```

classify and retrieve Complaints from a KB.

Embedding-Based Semantic Matching

The classification process begins by converting the complaint text into a high-

dimensional embedding vector using the *nomic-embed-text* model, which captures the semantic meaning of the complaint rather than relying solely on exact keyword matching. This embedding allows the system to understand the context and intent behind the user's input. The generated vector is then compared with precomputed embeddings stored in the *knowledge.json* dataset, where each government department is represented using structured information such as descriptions, keywords, synonyms, and example complaints. These elements are embedded and indexed beforehand, enabling efficient and fast similarity comparison during runtime. Based on this comparison, the system retrieves the most relevant department by identifying the closest semantic match, confirming that the classification is driven by contextual understanding rather than simple keyword-based matching.

Hybrid Scoring Mechanism

In addition to semantic similarity, there is a hybrid scoring strategy used in order to increase the accuracy of classifications. It includes a number of elements:

- Semantic similarity score (embedding-based)
- Keyword matching score
- Synonym detection score

All the mentioned components combined result in the hybrid score. Thus, having the highest score value – 0.578 – that exceeds the pre-defined threshold of 0.45, allows classifying the complaint in favor of the Health department.

This hybrid technique can be helpful in making sure that the system works properly even with less formal text data.

Confidence Threshold Validation

The system applies a minimum confidence threshold of **0.45** to determine whether the classification result is reliable. Since the computed score (**0.578**) is higher than the threshold, the system directly assigns the department without requiring further processing.

This mechanism helps in:

- Reducing incorrect classifications

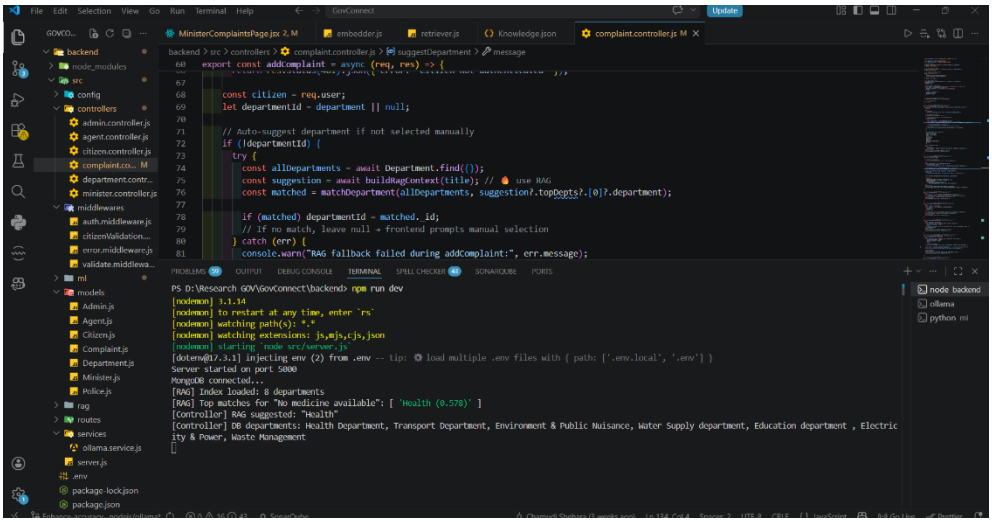
- Filtering weak matches
- Improving overall accuracy

LLM-Based Fallback Mechanism

However, when the hybrid score fails to meet the threshold requirement, the system activates the LLaMA 3 model to serve as a secondary decision-making module.The language model analyzes the background of the complaint and selects the most appropriate department from among the highly ranked results. This methodology guarantees that even those ambiguous complaints can be managed successfully.In the case of this example, although it doesn't activate the fallback option, its inclusion makes the system more robust.

Database Integration and Final Mapping

After determining the most suitable department for the customer's complaint, the system conducts an additional validation check by comparing the selected department with the database entries:



This ensures that:

- Only valid departments are allocated
- The classification is accurate based on the information in the database
- consistency is maintained

Upon successful validation, the system automatically allocates the correct department code to the complaint and proceeds to save it and move it forward in the process.

It is evident that in the case of GovConnect’s classification, the process takes

place automatically within the backend controller. In case a citizen makes a complaint and does not select the department manually, all available departments are extracted from the database. The next step entails the invocation of the method called buildRagContext(title) for the purpose of generating the department recommendation based on the classification pipeline, which is based on the RAG technique. This recommendation is verified in the database through matchDepartment(), and if a match occurs, an automatic assignment of the department ID takes place prior to saving the complaint.

In essence, what we have seen from this run is that the GovConnect classification engine is capable of comprehending complaints in natural language, doing more than just matching keywords to semantics, bringing in the right department, validating the data for consistency, and making AI-based decisions feed directly into the backend processes. This level of automation improves the efficiency of the complaints process and reduces the burden of government officials.

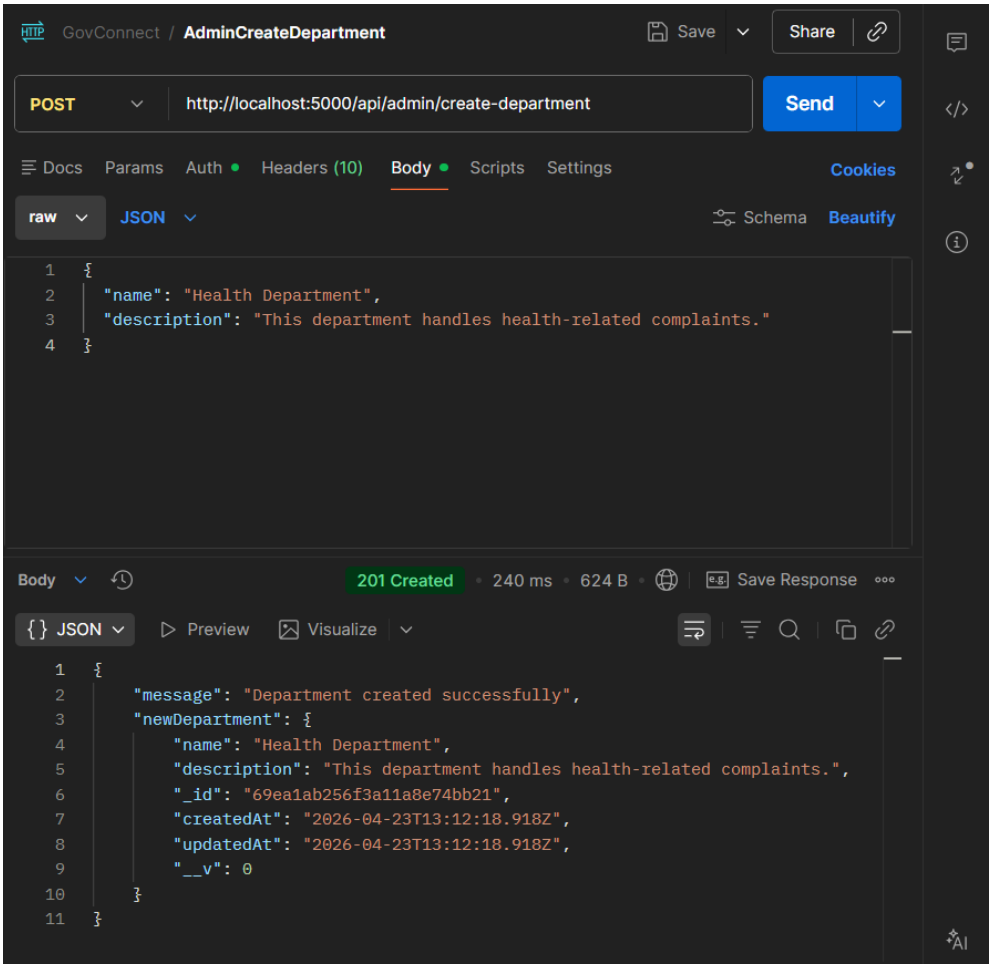
4.1.2. Department Creation Results

When testing how the system manages the creation of governmental organizational units, adding new departments was found to be effective when calling the admin API http://localhost:5000/api/admin/create-department.

For example, creating a department of Health resulted in a 201 Created result message and included essential information regarding department ID, department name, its description, and date and time of creation and last update.

This means that the backend can access the data stored in MongoDB, which ensures successful data management. At the same time, the test demonstrates that the system can scale easily, meaning that new departments can be easily added without modifying the codebase.

This feature has great significance from an architectural point of view, considering that governmental organization changes can occur over time.

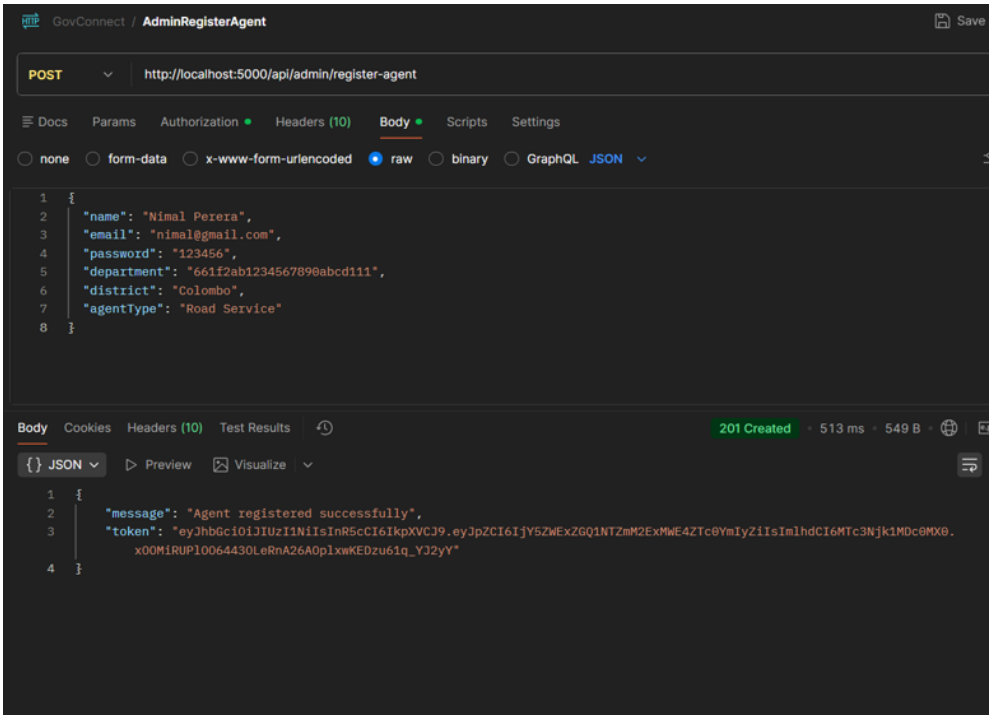


4.1.3 Agent Registration Results

Testing of agent registration was done by using the endpoint http://localhost:5000/api/admin/register-agent. The registration of agents was successful by incorporating data on department, district, and service types.

After registration, the system returns HTTP 201 created, together with the token for authentication purposes. This implies that this module incorporates proper handling of the hierarchy and linking of the agents to their departments.

One of the best features about this module is the use of districts in filtering. This is because the system assigns a local agent to any complaint from their district.



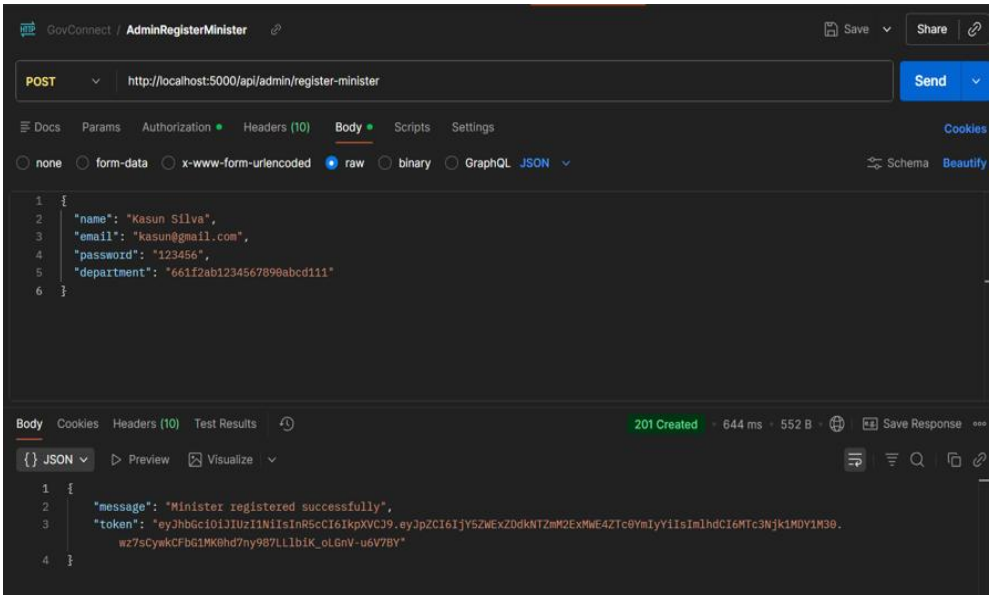
4.1.4 Minister Registration Results

Minister registration functionality was tested to ascertain whether it effectively handles the creation and authentication of users based on their roles. Calling the endpoint http://localhost:5000/api/admin/register-minister resulted in the creation of new minister accounts.

Upon successful registration, each account received a 201 status code and a JSON Web Token (JWT). It validates the fact that the authentication process is working correctly.

The system verifies the integrity of input data, securely stores credentials for all users, and assigns the proper role to everyone who logs into the system.JWT tokens offer session security and reduce cases of unauthorized access to sensitive functionalities in the system. It is important in government-related systems because of the need for data integrity and security.

Moreover, the system has strong role-based access control mechanisms, which limit account creation to administrative personnel only.



4.1.5 End-to-End Workflow Results

The comprehensive test for the GovConnect complaint management process was undertaken to verify that every aspect of the software operates as it should. First, the citizen reports a problem to the system, and it is automatically sorted using the RAG-based intelligent machine learning system. Once sorted, the complaint enters the database and shows up on the dashboard of the concerned minister. Thereafter, it is forwarded to an agent selected based on departmental affiliation and geographical region. Finally, the agent manages the complaint throughout its life cycle, regardless of whether it is pending, being resolved, or has been resolved. The findings from the test indicated that the whole process went through successfully without any problems within the software or data loss. Complaints were directed to appropriate departments, and agents were assigned according to set criteria. In summary, GovConnect provides an integrated and reliable mechanism for processing complaints.

4.1.6 System Performance Results

The performance of the system was analyzed through observing the time taken for the API backend to execute various critical functions. In the process of testing, the GovConnect platform showed that it can perform essential tasks within a short time. For example, the creation of a department required about 240 milliseconds, agent registration took approximately 513 milliseconds, minister registration needed 644 milliseconds, and logging in an administrator took only 226 milliseconds. It can be deduced from the above analysis that all the services executed behind the scenes are running smoothly under normal circumstances

4.1.7 Testing Accuracy of Complaint Classification System

The model's performance for classification purposes was evaluated using 39 real-life complaints. The model passed through the classification process based on the RAG approach.In the test process, the complaint text was transformed to an embedding vector form that was compared to knowledge base embeddings. A combined score for both semantic similarity and keyword matching was applied to achieve high performance, with a confidence threshold of 0.45.

Test Results: out of 39 test cases, 34 were classified correctly and 5 were incorrectly classified.

The accuracy was calculated as:

```
Accuracy  = (34/39)*100 = 87.18%
```

This outcome shows that the system classifies complaints with a high degree of accuracy.

4.2. Research Finding

From the tests conducted on the GovConnect model, we can highlight a few key takeaways regarding the performance of AI-assisted complaint classification and the efficiency of the integration. The assessment proves that the combination of RAG models with LLaMA 3 models is a viable and scalable option to facilitate automated complaint processing by the government agencies.

One of the key findings of this research was the high level of classification accuracy, which amounted to 87.18% with 34 classifications out of 39 complaints being correct. This highlights the strength of the hybrid approach that combines semantic similarity, keyword search, and synonyms. Moreover, the usage of embedding-based approaches enhances understanding and contextual interpretation of short and ungrammatical messages.

Another significant lesson to be learned is the confidence level of 0.45 set for ensuring that the classification can be considered credible. Since the system only allows classifications with higher confidence levels, errors and misclassification cases are eliminated, making the process very reliable. Another important aspect is the LLM fallback used in resolving ambiguous situations.

In practice, the system operated well: all complaints were successfully classified, mapped to relevant departments and integrated into workflows without any issues. The average API response time was kept under one second for critical operations.

Furthermore, it is evident that the workflow automation is crucial as it helps route the complaints to appropriate ministers and assign them to agents considering departments and districts..

Metric	Value
Total Test Cases	39
Correct Pedictions	34
Incorrect Predictions	5
Accuracy	87.18%

Table 4.2.1: Classification Performance Summary

4.3. Discussion

4.3.1. Effectiveness for AI-Based Classification

It is found that the AI tool, GovConnect, can effectively perform categorization of citizens' complaints using a hybrid model. This hybrid AI-based model not only looks at keyword match but also focuses on understanding the meaning of each complaint by considering its context. This

is why this model can provide accurate classification results regardless of how the complaints are phrased.

A high level of precision at 87.18% speaks volumes about the effectiveness of the model, as it was tested using natural language samples. In general, the combination of RAG and LLM approaches is quite efficient for the area of e-governance.

4.3.2 Impact of Hybrid Scoring Mechanism

With the hybrid approach, the classification of texts becomes more accurate as the classifier evaluates not only the semantic similarity but also the presence of keywords and synonyms in it. This approach is particularly advantageous in situations like these:

- Texts of complaints where the informal language is used
- Short texts
- Variations in wording

Introduction of a confidence threshold results in higher reliability of prediction, as the output contains only those matches that really belong there.

4.3.3 Role of LLM in Handling Ambiguity

One of the biggest strengths of the system is that in case the hybrid score does not perform well enough, the language model takes over and helps reason contextually and direct toward the best department.It really improves the performance of the system to deal with cases:

- Ambiguous complaints
- Multi-context issues
- Overlapping department categories

For example, the grievances related to any accident may concern both transportation and healthcare sectors. In this regard, the big language model solves the problem by focusing on the essence of the complaint.

4.3.4. Workflow Automation and System Integration

Moreover, while GovConnect not only classifies the grievances, it connects multiple functions in a seamless workflow. The study demonstrates that the complaints are automatically directed to the corresponding departments, evaluated by the ministers, and referred to the appropriate agents in accordance with districts and types of services provided.

Such automation reduces manual labor and increases the efficiency of the entire grievance management process. Every complaint is handled in a specific sequence from entry to processing.

4.3.5 System Performance and Scalability

As demonstrated from the results, all operations done through the API perform rapidly with most taking less than one second, proving the capability of the system to handle real-time requests.

Utilization of the MERN technology stack makes the system scalable since it can support additional users and complaints, and is also easy to expand in terms of features and functionalities due to its modular nature.

4.3.6 Limitations and Error Analysis

Despite high accuracy, the system was not able to correctly categorize five out of thirty-nine complaints in the test data set. These errors were attributed to:

- The existence of vague complaint descriptions
- Similar keywords for different departments
- Lack of explicit contextual information

This could happen when a complaint regarding an accident may be categorized under Health instead of Transport due to similarities in context.

4.3.7 Adaptability and Real-World Applicability

In addition, GovConnect demonstrates strong proficiency in processing different types of complaints ranging from informal messages to incomplete descriptions. The importance of the capability cannot be underestimated in practice since users are not likely to comply with any input forms strictly.

Finally, in contrast to traditional machine learning models, which require frequent retraining, the proposed approach remains light-weighted in terms of the process of learning. It is possible to change the knowledge base in order to process new complaints and governmental organizations.

4.3.8 Comparison with Traditional Systems

Unlike traditional machine learning algorithms, GovConnect does not undergo extensive training, validation, or constant retraining. Rather, the classification system operates using an expanding repository of knowledge coupled with semantic search capabilities that allow for dynamic adaptation. In this way, the classification system can accommodate new types of complaints without having to retrain the model itself.

5.CONCLUSION

GovConnect demonstrates the ways that Artificial Intelligence can improve the interactions between the government and its citizens by providing an intelligent system for handling complaints. The overall objective of our project was to design and implement an AI-driven system capable of automatically categorizing complaints from citizens into their proper departments, thereby integrating the technology into a comprehensive workflow process. This includes receiving complaints from citizens, forwarding them to the appropriate department, monitoring progress, and even allocating agents to address complaints.

From the results of the evaluation, it is clear that our proposed solution solves the weaknesses of the existing complaint management systems. Through the integration of RAG and LLaMA 3 models, the system understands the true essence of complaints and does not rely on keyword matching alone. Consequently, the classification accuracy of the system increases and enables the system to deal with natural language input. The classification accuracy of the system stands at 87.18%.

One of the other major insights gained from this project is the implementation of a classification system based on the combination of semantic, keyword,

and synonym matching. It adds robustness to the system as a whole, since it takes into consideration several perspectives of the language of the complaint text when classifying it. Moreover, the introduction of a threshold ensures a more confident decision-making process, as less confident predictions can be ignored. If there isn't sufficient confidence, then LLaMA 3 will be used to classify the complaint.

In addition to classification, the GovConnect system provides a complete end-to-end process for managing complaints. Concerns raised by citizens can be easily lodged using the user-friendly interface and will automatically be assigned to the appropriate departments. The ministers track the process of complaint management using specialized dashboards and assign tasks to agents depending on the department and geographic location. The agents document the entire process of each complaint as it progresses throughout its life cycle. Performance metrics confirm that the solution not only operates efficiently but also scales well. Functions performed by the backend, such as complaint categorization, department allocation, and user authentication, operate quickly, taking less than one second to complete. It suggests that the system can handle real-time transactions smoothly without any delays, thus making it feasible to deploy in large-scale environments where users and complaints are numerous.

Although the findings seem to be encouraging, the evaluation process also pointed out certain limitations. There have been several complaints that were erroneously classified by the model, indicating that the problem of ambiguity and multiple contexts is still not fully solved. When the problems relate to different spheres simultaneously – for instance, those related to transport and healthcare spheres – the similarity of the context may confuse the machine. In addition, the efficiency of the RAG-based classifier depends on the knowledge base.

In any case, these limitations do not hinder the performance of the system in any substantial way. On the contrary, these can be taken as the bases for making the system better in the future. The modular nature of the GovConnect

allows the database to be easily updated in order to accommodate new complaint types and new forms of governance. This is a key advantage over machine learning approaches that usually require retraining.

To conclude, it is possible to say that the GovConnect approach reveals how the introduction of AI-based approaches could contribute to enhancing the effectiveness of complaint handling procedures through increased speed, accuracy, and transparency. Integrating retrieval-augmented generation, large language models, and modern web technologies, the proposed model represents a scalable and intelligent solution capable of transferring the theoretical potential of e-governance into practical application. Instead of limiting itself to mere classification purposes, the model addresses all stages of the complaint handling process, thus serving as an integrated platform for e-governance.

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